MILITARY SPECIFICATION

TAPE, POLYESTER (FOR USE IN AMMUNITION)

1. SCOPE

1.1 This specification covers two types of polyester tape for use in ammunition (see 6.2).

1.2 Classification. Polyester tape shall be of the following types and classes, as specified (see 6.1).

Type I - Continuous Filament Polyester

	Weight, ounces/yard, max
Class 1	0.035
Class 2	0.05

Type II - Spun Polyester

Class 1	0.06
Class 2	0.08
Class 3	1.05

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-P-1134 - Packaging and Packing of Cotton and Cotton - Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics)

FSC: 8305

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STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191- Textile, Test Method

MILITARY

MIL-STD-105- Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109- Quality Assurance Terms and Definitions

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or requests for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-E105 - Probability Sampling of Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia Penn., 19103).

3. REQUIREMENTS

3.1 Material. The polyester tape shall be polyethylene terephthalate and shall meet the following requirements.

3.1.1 Type I. The Type I material shall be a tape made with continuous filament polyester yarns properly processed and of a suitable denier, filament, and twist to meet the requirements of this specification.

3.1.2 Type II. The Type II material shall be a tape made with spun polyester yarns properly processed and of a suitable denier, staple length and twist to meet the requirements of this specification.

3.1.3 Specific Gravity. The specific gravity of the tape shall be 1.38 $\pm$ .02 when determined as specified in 4.4.1.

3.1.4 Solubility. The tape shall be insoluble when determined as specified in 4.4.2.

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3.1.5 Physical properties. The physical properties for polyester tape shall comply with Table I when determined as specified in the applicable paragraph of Section 4.4.3 through 4.4.6.

TABLE I

	TYPE I				TYPE II					
Class:	1		2		1		2		3	
	min	max	min	max	min	max	min	max	min	max
Width, inches (in.)	.20	.28	.35	.50	.20	.28	.34	.43	.45	.57
Weight, ounces per yard (oz./yd.)	--	.035	--	.05	--	.06	--	.08	--	1.05
Breaking strength pounds per inch (lbs./in.) min	50		40		200		200		190	
Elongation, percent., max.	35		35		45		45		45	

Type I - No single break below 35 lbs/inch permitted.
Type II - No single break below 150 lbs/inch permitted.

3.1.6 Acidity or alkalinity

3.1.6.1 pH of water extract. The pH of the water extract of the tape shall be not less than 4.4 nor more than 9.7 when tested as specified in 4.4.7.

3.1.6.2 Acidity or alkalinity. The water extract of the tape shall show not more than 0.1 percent acidity, calculated as acetic acid, nor more than 0.1 percent alkalinity, calculated as sodium carbonate when tested as specified in 4.4.8.

3.1.7 Ash. The content of ash shall be 2.0 percent max when tested as specified in 4.4.9.

3.2 First article testing. This specification makes provisions for first article testing. Requirements for the submission of first article sample by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein.

4.1.1 Submission of Product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted.

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agents authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission.-Prior to the start of regular production the contractor shall submit a first article sample (see 6.1) to a Government approved facility as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2 The first article sample shall consist of five (5) rolls of tape in accordance with instructions issued by the Contracting Officer. All samples submitted shall have been produced by the contractor using the same production processes, procedures, and equipment as will be used in fulfilling the contract. All materials,

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including packaging and packing shall be obtained from the same source of supply as will be used in regular production. The sample shall be accompanied by certificates of analysis. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect production uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspections to be performed. The sample will be subjected by the Government to any or all of the examinations or tests specified in 4.4 of this specification.

4.2.3 Rejection. If any sample fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a sample to comply with any of the stated requirements.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process using the same materials and methods, with the same specifications and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specifications, not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted

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for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements.

4.3.2 Examination of the end item. Examination of the end item shall be in accordance with 4.3.2.1 through 4.3.2.3.

4.3.2.1 Yard-by-yard examination. Each roll in the sample shall be examined on the face side. When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards, only the first 100 yards per roll shall be examined. The sample size shall be in accordance with the following:

Lot size (yards)	Sample Size (rolls)1/
3200 or less	8
3201 up to and including 10,000	13
10,000 and over	20

1/No more than one roll shall be taken from any shipping container unless the number of shipping containers in the lot is less than the required number of rolls, in which case, all shipping containers shall be present in the sample. All defects as defined in Section III of FED-STD-4, which are clearly noticeable at normal inspection distances (3 feet) shall be scored and assigned demerit points as listed in 4.3.2.2. No linear year (increment of 1 yard on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than 4 points. The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceeds 35 points. The lot shall be unacceptable if the points per 100 square yards of two or more individual rolls exceed 53.0 points. If one roll exceeds 45.0 points per 100 square yards, a second sample equal in size to the first sample shall be examined only for individual roll quality examination. The lot shall be unacceptable if one or more rolls in the second sample exceeds 45.0 points per 100 square yards. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample X 3600}}{\text{Contracted width of cloth(inches) X total yards inspected.}} = \text{points per 100 square yards}$$

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4.3.2.2 Demerit points. Demerit points shall be assigned as follows:

For defects up to and including 3 inches in any direction - one point
For defects exceeding 3 inches, but not exceeding 6 inches in any direction - two points
For defects exceeding 6 inches, but not exceeding 9 inches in any direction - three points
For defects exceeding 9 inches in any direction - four points

The following defects, when present, shall be scored four points for each yard in which they occur:

Hole, cut, tear or smash
Overall uncleanness
Objectional odor
Cloudy, mottled or streaky

4.3.2.3 Examination for length

4.3.2.3.1 Individual roll. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.3.2.3.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of the rolls in the sample is less than the total of the lengths marked on the tickets. The rolls examined shall be those selected for the examination of individual rolls.

4.3.3 Sampling. Sampling shall be in accordance with ASTM Method E105. One container (roll) shall be sampled from each lot. Five yards of material shall be subjected to the required inspections. No inspection shall be performed on the first five yards of material from the roll selected. If the sample fails to meet any of the requirements, the lot shall be rejected. The classification and code numbers for the tests are given below.

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Category	Defect	Code No. (see 6.2)
Specific Gravity	Major	01001
Solubility	Major	02001
Width	Major	03001
Weight	Major	04001
Breaking Strength	Major	05001
Elongation	Major	06001
pH	Major	07001
Acidity or Alkalinity	Major	08001
Ash	Major	09001

4.4 Test methods

4.4.1 Specific Gravity. A sample of the tape shall be placed in a fat-free extraction thimble (Whatman single thickness) to prevent escape of sample. Thimble with sample is extracted in a Soxhlet extractor, using chloroform, for at least 20 cycles. After extraction the sample shall be thoroughly dried. A liquid of specific gravity 1.36 and liquid of specific gravity 1.40 shall be prepared by mixing the proper proportions of either xylene or benzene with perchlorethylene and checking the specific gravity with a hydrometer. The extracted sample for specific gravity determination shall be prepared by tying a knot in the polyester yarn and cutting the yarn on each side of the knot. Boil the sample for 5 minutes in either benzene or xylene. Remove the sample with a tweezer and place below the surface of the liquid of known specific gravity in a test tube. The polyester fiber shall sink to the bottom of the tube when placed in liquid of specific gravity 1.36 and float in the liquid of specific gravity 1.40.

4.4.2 Solubility. Place separate samples of the tape into individual beakers containing boiling acetone, 70 percent sulfuric acid, 5 percent sodium hydroxide and boiling methylene chloride. Allow tape to remain in contact with the solvent for ten minutes with intermittent stirring. Visually check if the solvent attacked or dissolved the tape.

4.4.3 Width. Determine the width of the tape in accordance with Method 5020 of FED-STD-191.

4.4.4 Weight. Determine the weight of the tape in accordance with Method 5040 of FED-STD-191.

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4.4.5 Breaking strength. Determine the breaking strength of the tape in accordance with Method 5100 of FED-STD-191.

4.4.6 Elongation. The elongation of the tape shall be obtained when the breaking strength is determined by means of a suitable autographic recording device on the testing machine. An initial load of six ounces shall be placed on the specimen before gripping it in the lower jaws of the machine. The elongation shall be the average of the results obtained for five specimens. It shall be calculated from the start of the line as shown on the graphic record and expressed as the percentage increases in length.

4.4.7 pH of water extract. Determine the pH of the water extract in accordance with Method 2811 of FED-STD-191.

4.4.8 Acidity or alkalinity. Decant the water extract obtained in the determination of pH into a 250 ml beaker forcing as much absorbed water as possible out of the tape by pressing the tape against the side of the beaker with the aid of a glass stirring rod. If the pH of the water extract is 7.0 or greater, reddish tint.

Calculate any alkalinity to percent sodium carbonate as follows:

$$\text{Percent sodium carbonate} = \frac{5.3\ VN}{W}$$

Where:

V = ml of hydrochloric acid solution required
N = normality of the hydrochloric acid solution
W = weight of sample

If the pH of the water extract is 6.9 or less, titrate the solution with approximately 0.1N sodium hydroxide solution of known normality using phenolphthalein indicator to the first permanent pink color. Claculate any acidity to percent acetic acid as follows:

$$\text{Percent of acetic acid} = \frac{6.0\ VN}{W}$$

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Where:

V = ml of hydrochloric acid solution required
N = normality of the hydrochloric acid solution
W = weight of the sample

4.4.9 Ash: Cut parts from each of the pieces comprising the sample so as to obtain a total of approximately two gm. and transfer to a tared platinum or porcelain crucible and moisten with concentrated nitric acid. Heat the crucible on a steam bath for one hour and then heat carefully over a flame so as to avoid any loss by spattering. When the contents of the crucible have been charred, heat to dull redness until all carbonaceous matter has been burned off. Cool the crucible in a desiccator and weigh. Calculate the gain in weight to percent ash in the sample.

5. PREPARATION FOR DELIVERY

5.1 Level C - Packaging, packing and marking - Packaging packing and marking shall be in accordance with PPP-P-1134.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification
b. Data cards shall be prepared for each lot in accordance with MIL-STD-1167.
c. Provisions for submission of first article samples.
d. Type and class

6.2 Inspection code numbers. The five digit code numbers assigned to the inspection herein are to facilitate future data collection and analysis by the Government.

6.3 Intended use. The polyester tape is intended for use in the 155MM, XM203 and XM188 Propelling Charges.

Custodian:
ARMY-MU

Preparing Activity:
ARMY-MU

Project Number: 8305-A390

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